

Hacker Holidays 2026 – Day 4

Packed Light

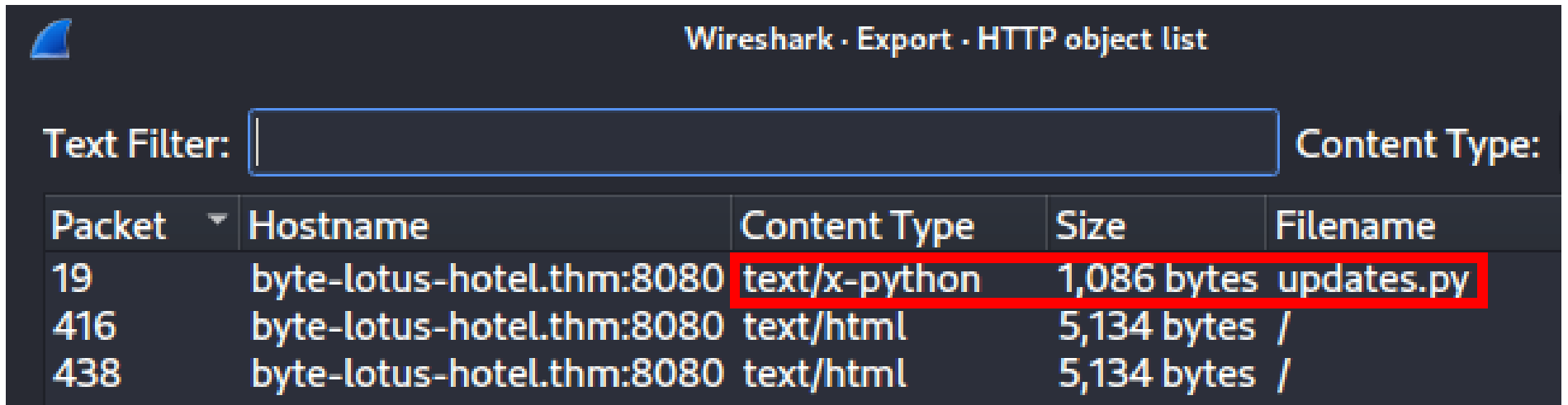
TODAY'S ITINERARY

- ☐ Analyze the provided capture for a covert communication channel.
- ☐ Identify where the exfiltrated data is being hidden and reassemble it.
- ☐ Decode the recovered data and submit the flag.

In this challenge for the Hacker Holidays, we're asked to analyze a network packet capture file to find an encrypted

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The image shows a screenshot of the Wireshark 'Export - HTTP object list' window. At the top, there is a 'Text Filter:' input field and a 'Content Type:' label. Below these is a table with five columns: 'Packet', 'Hostname', 'Content Type', 'Size', and 'Filename'. The table contains three rows of data. The first row, for packet 19, is highlighted with a red border. The 'Content Type' and 'Filename' cells in this row are also highlighted with a red background. The other two rows show HTML content.

Packet	Hostname	Content Type	Size	Filename
19	byte-lotus-hotel.thm:8080	text/x-python	1,086 bytes	updates.py
416	byte-lotus-hotel.thm:8080	text/html	5,134 bytes	/
438	byte-lotus-hotel.thm:8080	text/html	5,134 bytes	/

When we look at the **File** → **Export Objects** → **HTTP** option, we see that there's a Python script included in the packets, named **update.py**

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```
encrypted = xor(raw_bytes, getkey().encode('utf-8'))  
b64_string = base64.b64encode(encrypted).decode('utf-8')
```

When we inspect the script, it describes a method for encrypting a message with a combination of XOR encryption and base64 encoding

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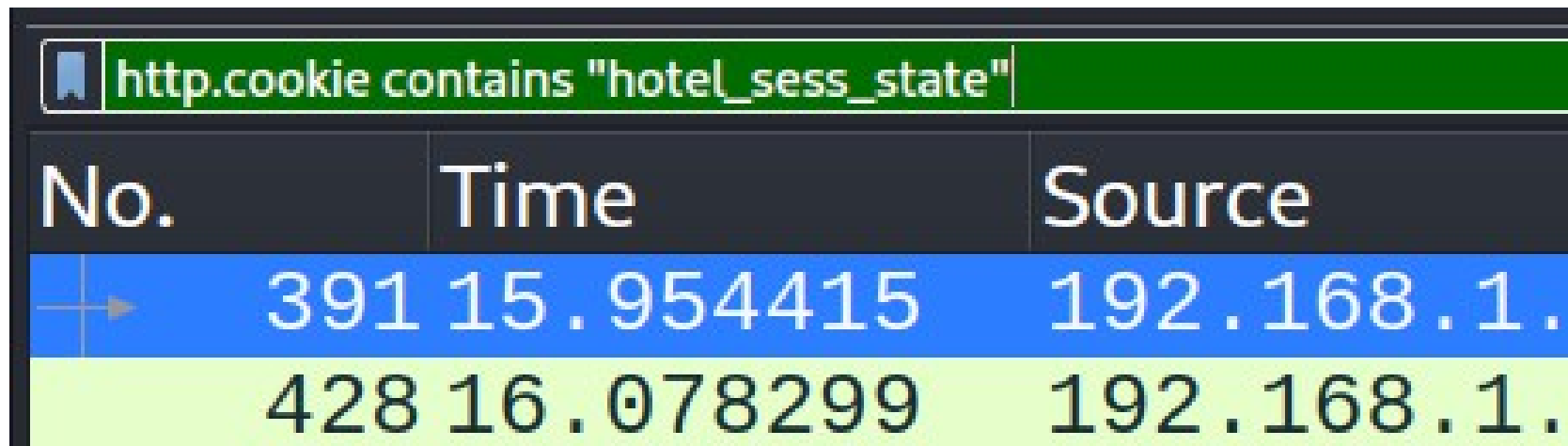
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```
headers = {  
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) ByteLotusClier  
    "Cookie": f"hotel_sess_state={b64_string}"  
}
```

The characters are hidden in the **hotel_sess_state** cookie in a series of HTTP responses

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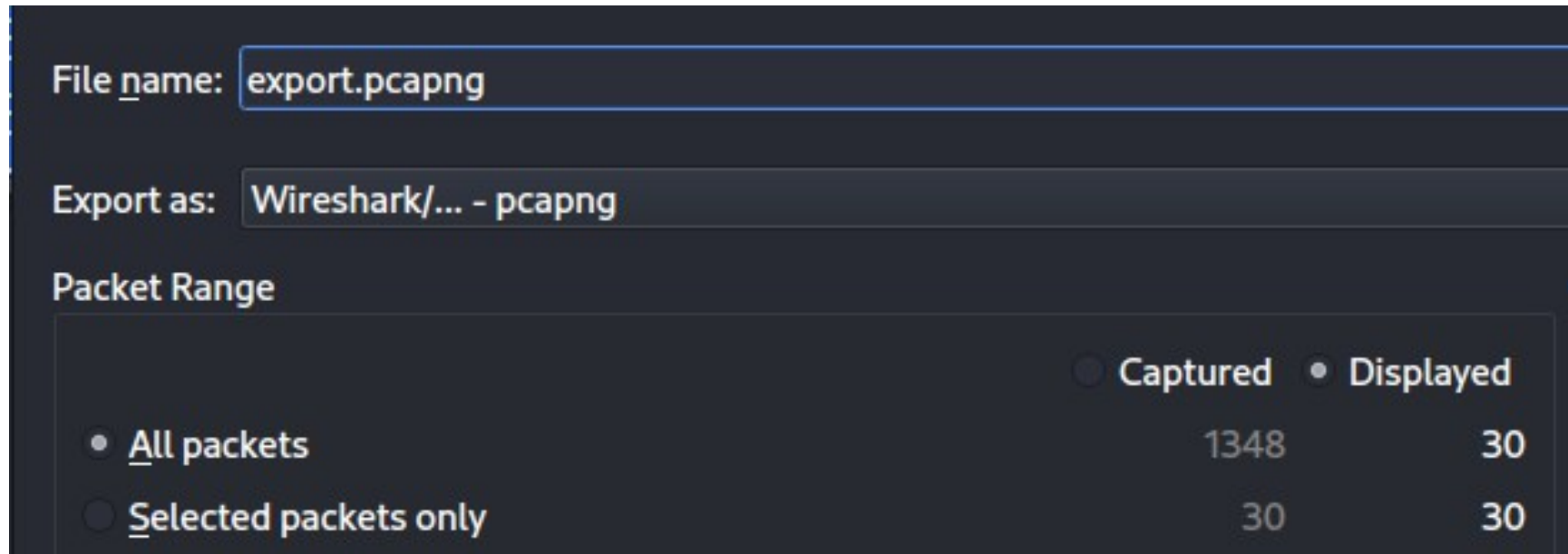
The image shows a Wireshark interface. At the top, a green filter bar contains the text 'http.cookie contains "hotel_sess_state"'. Below this is a table of captured packets. The first packet is highlighted in blue, and the second is highlighted in light green.

No.	Time	Source
391	15.954415	192.168.1.
428	16.078299	192.168.1.

We can locate all of the packets that contain the hotel_sess_state cookie by using the Wireshark filter `http.cookie contains "hotel_sess_state"`

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Then export all of these packets to a separate file using the **File** → **Export Specified Packets** option

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```
└─$ strings export.pcapng | grep Cookie | cut -d "=" -f 2-  
HA=  
AA=  
BQ=
```

We can use a series of Linux commands to extract only the base64 strings, and output them to a file

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```
Input                                     length: 149
                                       lines: 30
...
Fw==
eA==
Pg==
LQ==
Gg==
Fw==
MQ==
eA==
PQ==
NQ==
.....
Output                                   time: 15ms
                                       length: 30
                                       lines: 1
THM{V0nd_1s_watching_0ver_y0u}
```

Finally, we can decrypt the message using a combination of base64 decoding and XOR decryption, using CyberChef